THE CASE FOR FURTHER EXPLORING THE LINK BETWEEN WOMEN'S BRAIN HEALTH AND THE ECONOMY

Evidence from informal and academic datasets, 22 July 2026

BRAIN HEALTH AS AN ECONOMIC INPUT

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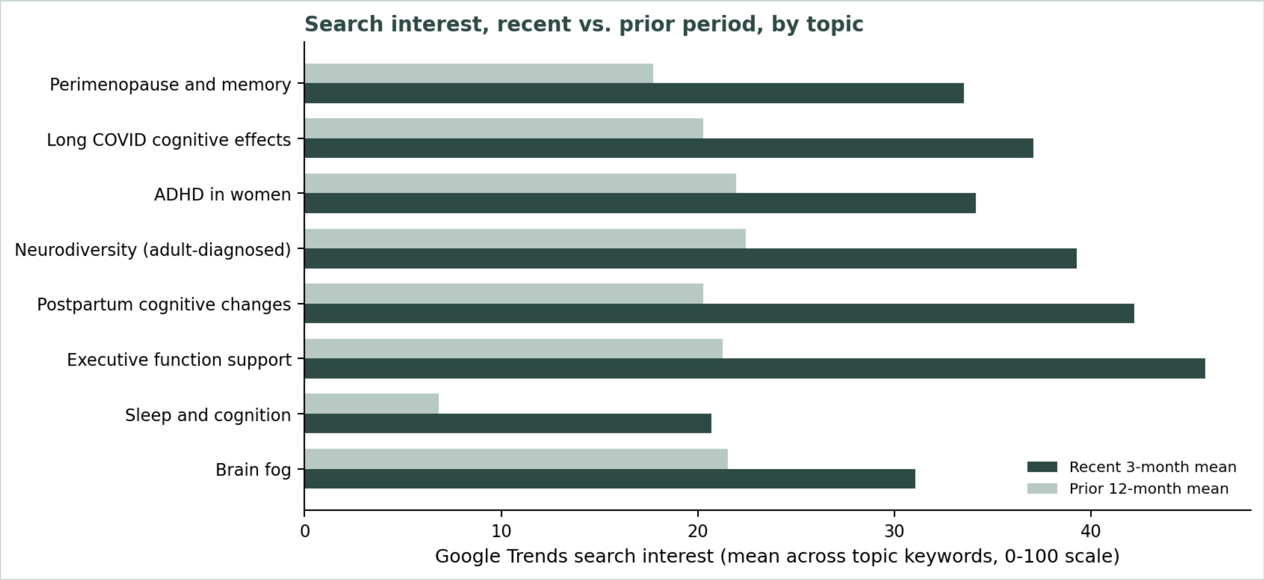
Interest in women's brain health and its correlation with labour have soared since 2021: across eight related topics, public search interest rose by an average of 115% and academic publishing on the same topics grew by an average of 65%, comparing a recent snapshot against the prior period.

A Google Trends keyword combination search revealed particularly sharp growth for specific terms: "postpartum memory" search interest is up 251.5%, "sleep deprivation cognition" up 235.7%, "post covid cognitive" up 213.7%, and "perimenopause memory" up 191.0%, each comparing the recent 3-month mean against the prior 12-month mean.

The graphic below shows that evolution in interest and users' search, comparing the recent 3-month mean against the prior 12-month mean for each topic.

FIGURE 1 GOOGLE TRENDS SEARCH INTEREST, RECENT VS. PRIOR PERIOD, BY TOPIC



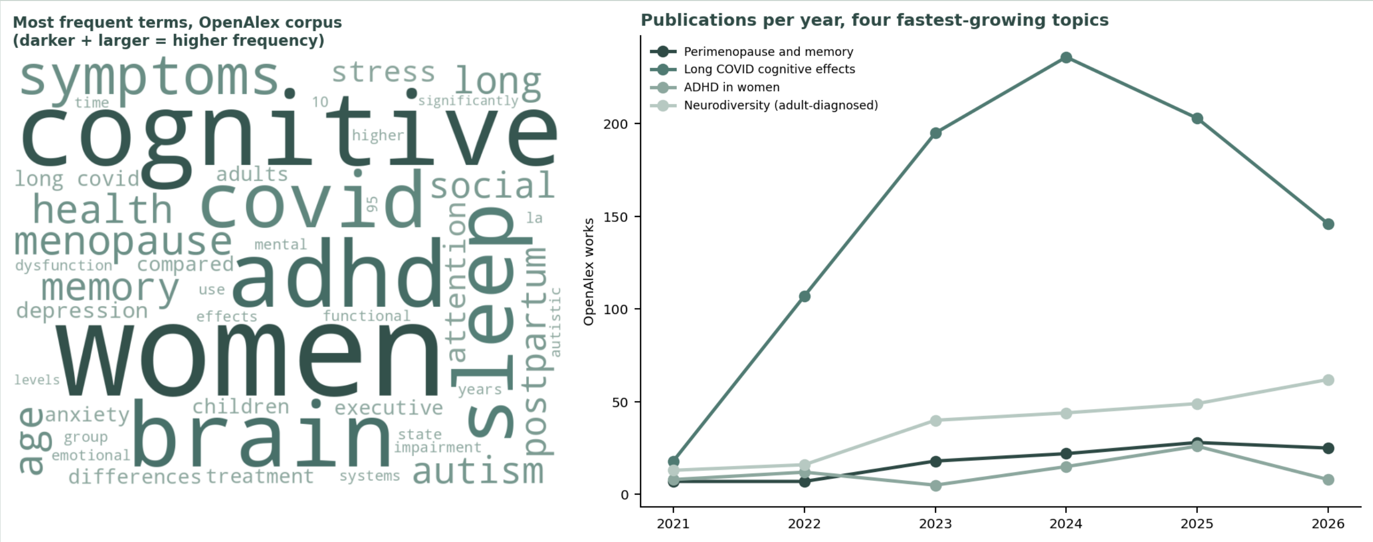
Source: own analysis, Google Trends (pytrends), 22 July 2026.

ACADEMIC PUBLISHING EVIDENCE: THE OPENALEX QUERY

Continued

An OpenAlex query also revealed a matching rise in academic output: publications on Long Covid cognitive effects grew from 18 in 2021 to a peak of 236 in 2024; work on neurodiversity in adults rose every year, from 13 in 2021 to 62 already in the partial 2026 count; and perimenopause-and-memory publications more than tripled, from 7 in 2021 to 25 so far in 2026. The word cloud and publication trend below are drawn from that same OpenAlex corpus.

FIGURE 2 MOST FREQUENT TERMS IN THE OPENALEX CORPUS, AND PUBLICATION GROWTH FOR THE FOUR FASTEST-GROWING TOPICS



Source: own analysis, OpenAlex works corpus (word cloud) and OpenAlex Works API by year (trend lines), 22 July 2026. 2026 figures are partial-year.

PREMISE: WHY THIS PROPOSAL, AND WHAT IT TESTS

Continued

POLICY QUESTION

Does discourse from public and unconventional datasets amplify academic evidence of female brain health being economically material?

OECD has already made the case that women's brain health is economically material: its New Approaches for Economic Challenges unit has run Brain Capital, a programme treating brain health as a core economic asset, since 2021, including work on sex and gender differences in neurological outcomes[1]. What is not yet tested is whether public, unconventional data, search interest, publishing activity, amplifies that case: gives it earlier, cheaper, independently observable evidence than the institutional argument alone provides.

The labour-market cost that argument points to is already measured: menopausal symptoms are linked to higher sick leave and lower work productivity[2], adult ADHD to comparably large productivity loss[3], and Long Covid to higher odds of leaving employment the longer symptoms persist[4]. If unconventional public data tracks the same conditions, it gives policymakers a faster, cheaper signal on top of the institutional case, not a competing one.

This proposal tests a narrower, answerable version of that question: for eight topics spanning the female lifespan, perimenopause and memory, adult ADHD in women, long COVID cognitive effects and others, is public search interest and academic publishing growing fast enough to count as that kind of evidence. This methodology uses two free data sources, Google Trends and OpenAlex. It does not yet run the medical-recognition check itself, against MeSH, the step that would show whether a topic already has a formal classification.

Working hypothesis: **these eight topics are gaining public and research attention faster than they are gaining formal medical recognition. The eight were chosen because each names a phase of the female lifespan or a related cognitive condition already raised in OECD's Brain Capital seminars, not because they scored highest in a broader keyword sweep.**

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TOPICS WITH POSITIVE EMERGENCE

4
TOPICS SCORING ABOVE 8,000

OBJECTIVE, THIS ROUND

Confirm, from this single 22 July 2026 data pull, whether at least half of the eight topics show positive growth on both signals, enough to justify running the MeSH comparison next round and serve as basis for further research.

LIMITATIONS

- No comparison against existing medical terminology yet.
- No combined priority score, only the emergence half.
- No trend line beyond the years shown here, one collection round only.

(1) Eyre et al. (2021), *Neuron*. (2) O'Neill, Jones & Reid (2023), *Occupational Medicine*. (3) Joseph et al. (2019), *Journal of Attention Disorders*. (4) Ayoubkhani et al. (2024), *European Journal of Public Health*. Full citations on page 5.

WHAT THE TWO DATA SOURCES SHOW

Continued

ALL EIGHT TOPICS ARE GROWING; FOUR STAND OUT

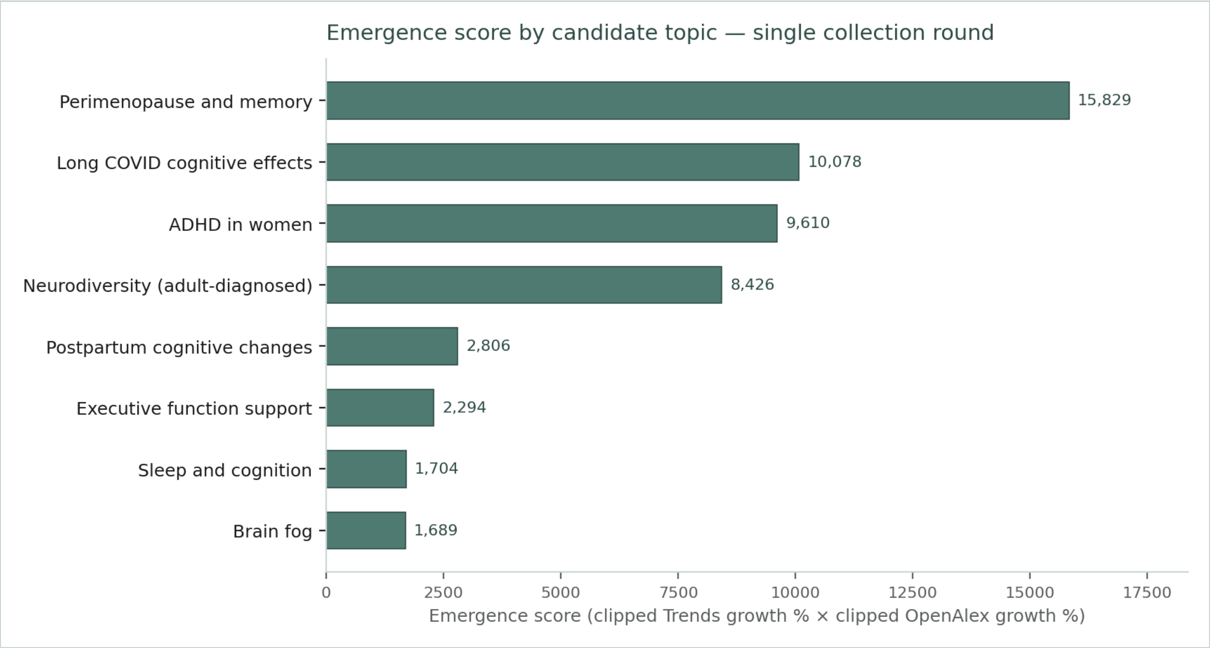
Perimenopause and memory, long COVID cognitive effects, ADHD in women and neurodiversity (adult-diagnosed) combine rising Google Trends search interest with rising OpenAlex publication counts at the same time, exactly the kind of convergence the emergence-score formula was designed to reward rather than either signal rising on its own. The remaining four topics are still growing on both signals, just less sharply.

Emergence score is calculated exactly as pre-registered: clipped Google Trends growth (recent 3-month mean vs. prior 12-month mean) multiplied by clipped OpenAlex publication growth (trailing 3-year works vs. prior 3-year works from a 6-year series), both terms floored at zero before multiplying[1]. No window or formula choice was adjusted after seeing these numbers.

EXECUTIVE FUNCTION SUPPORT IS A KNOWN OUTLIER

Its OpenAlex counts (80,067 prior-period works, 97,559 recent) are an order of magnitude larger than every other topic's, because both of its keyword variants ("executive function", "executive function support") are generic clinical terms without a women's-health qualifier. That risk was written down as a limitation before the data was collected, not adjusted after seeing the numbers.

FIGURE 1 EMERGENCE SCORE BY CANDIDATE TOPIC



Source: own analysis, Google Trends (pytrends) / OpenAlex Works API, 22 July 2026.

(1) Both growth terms are clipped at zero before multiplying, a rule fixed before this data was collected, so that two co-declining signals cannot produce a false-positive emergence score.

EVIDENCE TABLE AND WHAT HAPPENS NEXT

Continued

EVIDENCE SUMMARY

EMERGENCE COMPONENTS BY TOPIC, RANKED BY EMERGENCE SCORE

Topic	Trends growth %	OpenAlex growth %	Emergence score
Perimenopause and memory	117.8	134.4	15,829
Long COVID cognitive effects	121.7	82.8	10,078
ADHD in women	100.1	96.0	9,610
Neurodiversity (adult-diagnosed)	67.6	124.6	8,426
Postpartum cognitive changes	133.3	21.1	2,806
Executive function support	105.0	21.8	2,294
Sleep and cognition	217.9	7.8	1,704
Brain fog	53.1	31.8	1,689

EVIDENCE TABLE AND WHAT HAPPENS NEXT

Continued

READING THE RANKING CORRECTLY

Sleep and cognition has the highest Trends growth (217.9%) but ranks near the bottom on emergence because its OpenAlex growth is flat (7.8%): the multiplicative formula rewards convergence across both signals over strength on either one alone. The same logic explains why postpartum cognitive changes, the highest Trends growth of all eight at 133.3%, still ranks fifth.

WHAT'S STILL MISSING

The medical-recognition check has not been run yet. This page compares public search interest against academic publishing only, not yet against MeSH, the US National Library of Medicine's controlled vocabulary, which would show whether a topic already has formal recognition. Adding that comparison would produce a single priority score per topic; without it, these numbers describe attention and publishing activity only.

FUTURE STEPS

The initial analysis has identified the most promising topics. As a next step, it may be valuable to conduct additional research into MeSH (Medical Subject Headings) integration and explore opportunities to further enrich the prototype. This could include evaluating how MeSH terminology mapping may improve topic classification, strengthen consistency across analyses, and support future scalability. In parallel, expanding the prototype's functionality could help assess its robustness and potential value beyond the current scope. Further investigation in these areas would provide greater confidence in the approach and help identify opportunities for enhancement before proceeding to a a broader research exercise.

EVIDENCE TABLE AND WHAT HAPPENS NEXT

Continued

REFERENCES

Academic — Eyre et al. (2021). Building brain capital. *Neuron*, 109(9).

O'Neill, Jones & Reid (2023). Impact of menopausal symptoms on work and careers. *Occupational Medicine*, 73(6).

Joseph et al. (2019). Health-related quality of life and work productivity of adults with ADHD. *Journal of Attention Disorders*, 23(12).

Ayoubkhani et al. (2024). Employment outcomes of people with Long Covid symptoms. *European Journal of Public Health*, 34(3).

Institutional — OECD NAEC Unit, Neuroscience-inspired Policy Initiative, Brain Capital programme.

Data sources — Google Trends (pytrends); OpenAlex Works API, 22 July 2026.

BRAIN HEALTH AS AN ECONOMIC INPUT · POC draft research proposal, 22 July 2026.

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